

Ramneek Kaur

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Mohali, Punjab, India

INTRODUCTION

- Aspiring Software & AI/ML Engineer with strong foundations in C/C++, Python, Data Science, and Machine Learning
- Experienced in developing applications across the full Software Development Life Cycle (SDLC) — from requirements and design to coding, testing, and deployment
- Proficient in OS fundamentals including process management, memory management, and inter-process communication on Windows and Linux platforms
- Skilled in building modular, reusable software components and integrating them into larger systems and subsystems
- Passionate about solving real-world problems with clean, maintainable code and a strong consumer-needs-first mindset

PROJECTS

RAG over Audio

Aug 2025

- Designed and implemented a full SDLC pipeline — requirements gathering, architecture design, coding, testing, and deployment — for an AI-powered question answering system over audio
- Implemented audio transcription using AssemblyAI and semantic retrieval using the FAISS vector database
- Integrated LangChain with Google Gemini 1.5 Flash to generate context-aware responses; exposed functionality via modular, reusable components
- Built an interactive Streamlit interface for real-time audio query processing

Conversational AI Chatbot

Aug 2025

- Developed an AI chatbot following structured SDLC phases — design, implementation, integration testing, and iterative improvement
- Built with Python and Google Gemini API; implemented session-state-based conversation memory for context-aware interactions
- Designed clean, modular code architecture to allow easy extension and integration of new conversational features

Volume Controller using Hand Gesture

Jan 2025

- Developed a C++ and Python computer vision application to control system volume using hand gestures on the Windows platform
- Implemented hand tracking with OpenCV and MediaPipe; integrated with Windows OS audio APIs via pycaw for system-level volume control
- Applied OS fundamentals — system calls and hardware abstraction — to interface application logic with the OS audio subsystem
- Enabled touchless interaction suitable for accessibility-oriented and embedded applications

AI Web Scraper

Jan 2025

- Built a modular Python web scraping system following SDLC best practices — component design, coding, unit testing, and maintenance
- Combined BeautifulSoup and custom data transformation modules into a reusable pipeline for structured data extraction to CSV and Excel
- Implemented robust issue handling and error recovery to support long-running, reliable data collection jobs

EDUCATION

Chandigarh University

2023 – Present

B.E. Computer Science Engineering (AI & ML Honors) — CGPA: 8.82

Ryan International School, Chandigarh

2023

Class XII, CBSE — 91.4%

St. Xavier's High School, Mohali

2021

Class X, ICSE — 96.4%

SKILLS

Programming: C, C++, Python, JavaScript, SQL

Software Engineering: SDLC, Modular Design, Component Integration, Code Review, Issue Investigation, Maintenance & Complaints Handling, REST APIs

OS & Systems: Windows (primary), Linux, Process Management, Memory Management, IPC, System Calls, Hardware Abstraction

AI & ML: Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Feature Engineering

Libraries & Frameworks: OpenCV, MediaPipe, LangChain, FAISS, Pandas, NumPy, Scikit-learn, Streamlit, Node.js, Express.js, React

Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Postman

CERTIFICATIONS

- Microsoft Certified — Azure AI Fundamentals
- NVIDIA — Foundations of Deep Learning
- Google Cloud — Introduction to Generative AI
- Intel India — AI for Entrepreneurship

EXTRACURRICULAR ACTIVITIES

- British Airways Data Science Job Simulation (Forage) — end-to-end SDLC simulation including data analysis, modelling, and presentation
- Deloitte Australia Data Analytics Job Simulation (Forage) — practical experience in data processing, consumer-needs analysis, and reporting
- Active participant in AI and Data Science hackathons — collaborative team settings with fast-paced problem solving and delivery